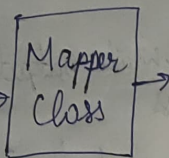


* It controls how keys are distributed among different reducers.

Partitioner

1) Mapper Class

apple ball apple cat
ball cat dog egg
nut orange zebra



(apple, 1) (ball, 1)
(ball, 1) (cat, 1)
(apple, 1) (dog, 1)
(cat, 1) (egg, 1)
(nut, 1) (orange, 1)
(zebra, 1)

* Reads a line of text.

* Splits the text into words.

* Emits each word as key with 1 as value.

2) Partitioner class (Alphabet Partitioner)

* Partitions words based on the first letter.

Word starts with

* a - m $\Rightarrow$ it goes to reducer 0

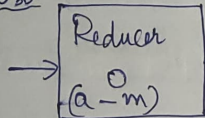
Word starts with

n - z

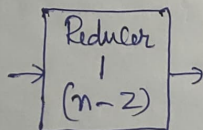
$\Rightarrow$ it goes to reducer 1

3) Reducer Class

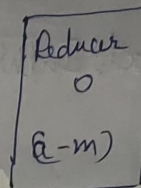
(apple, [1, 1])
(ball, [1, 1])
(cat, [1, 1])



apple 2
ball 2
cat 2

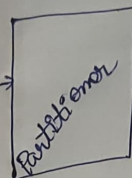


(apple, 1) (ball, 1)
(ball, 1) (cat, 1)
(apple, 1) (dog, 1)
(cat, 1) (egg, 1)

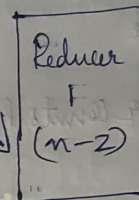


apple 2
ball 2
cat 2
dog 1
egg 1

(apple, [1, 1]) (egg, [1, 1])
(ball, [1, 1])
(cat, [1, 1])
(dog, [1, 1])



(nut, 1)
(orange, 1)
(zebra, 1)



nut 1
orange 1
zebra 1

(nut, [1, 1])
(orange, [1, 1])
(zebra, [1, 1])

Final output (Stored in HDFS)

apple 2
ball 2
cat 2
dog 1
egg 1
nut 1
orange 1
zebra 1

Partitioner < Text, IntWritable >

1) Text :-

- * Key type

- * Ex:- words like "apple", "banana" etc.

2) IntWritable :-

- * Value type

- * Ex:- integer counts like 1, 2, 3 --

getPartition (Text Key, IntWritable Value,
int numReduceTasks)

1) Text Key :-

- * Word (Ex:- "apple", "banana")

2) IntWritable Value :-

- * Count associated with the word (eg: 1)

3) int numReduceTasks :-

- * Total no. of reducers set in the job.

Char firstLetter = Character.toLowerCase (key.toString().
CharAt (0));

- * This extracts first letter of the word and converts it to lowercase.

* `charAt(0)` → get the character at index 0.
(the first character)

* `Character` → built-in Java class in the `java.lang` package that provides utility methods for working with character (`char` type).

* `StringTokenizer` → A Java class used to break a string into smaller parts (tokens), usually words. (default value is whitespace)